MILITARY SPECIFICATION SHEET

ELECTRON TUBE, CATHODE RAY

TYPE 7BP7A

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Magnetic deflection and focus

PIN CONNECTIONS AND DIMENSIONS: See figure 1

ABSOLUTE RATINGS:

Parameter:	Ef	Ec1	Ec2	Eb1	Ehk	Rg1	Alt
Unit:	V	Vdc	Vdc	Vdc	Vdc	Meg	ft
Maximum:	6.9	0	770	8,800	-125	Note 1	10,000
Minimum:	5.7	-180	---	4,000	---	---	---
TEST CONDITIONS:	6.3	Adj	250	4,000	---	---	---

GENERAL:

Qualification - Required (see note 2)

(B) denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	LIMITS		UNIT
					MINIMUM	MAXIMUM	
	Qualification inspection						
(B) 5101	Neck and bulb alinement (magnetic types)	-		---	---	---	---
(B) 5101	Face tilt	-		---	---	---	---
(B) 5216	Cathode illumination	-		---	---	---	---
1331	Direct-interelectrode capacitance	-	Cathode to all Control grid to all Grid No. 2 to all	Ck Cg1 Cg2	--- --- ---	10 11 12	pF pF pF
1141	Pressure (implosion)	-		---	---	---	---
5111	Vibration	-		Width	---	2	mm
(B) 1216	Base material insulating quality	-	Zone 5, minimum	---	---	---	---
	Quality conformance inspection, part 1						
5201	Voltage breakdown	-		---	---	---	---
5201	Voltage breakdown (magnetic types)	-		---	---	---	---
5206	Gas ratio	4		Gr	---	0.25	---
5101	Neck straightness	-		---	---	---	---
5106	Bulb, screen, and faceplate quality	-		---	---	---	---
(B) 5223	Modulation	-	Ib = 200 µAdc	ΔEc1	---	38	Vdc
5231	Spot position (magnetic deflection)	-		---	---	12	mm
5236	Zero bias anode current (magnetic deflection)	-		---	---	---	---
5241	Grid cutoff voltage	-		Ec1	-25	-70	Vdc
5251	Grid No. 1 leakage current	-		---	---	---	---
---	Aperture alinement	3		Distance	---	6	mm
	Quality conformance inspection, part 2						
1301	Heater current	-		If	540	660	mA
(B) 5201	Electrode current (grid No. 2)	-	Ec1 = 0	Ic2	---	50	µAdc
5101	Side terminal and base alinement	-	Pin No. 5	---	---	---	---
5216	Stray light emission (conventional types)	-	Eb = 8,800 Vdc; Ec2 = 770 Vdc	---	---	---	---
5221	Screens	-		---	---	---	---
5226	Line width A	-	Ib = 200 µAdc	Width	---	0.65	mm
5226	Line width C	-	Ib = 200 µAdc	Width	---	0.75	mm
5246	Focusing ampere turns	-	Ib = 200 µAdc; D = 2.75 inches	AT	337	459	---

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or external flashover

METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	LIMITS		UNIT
					MINIMUM	MAXIMUM	
	Quality conformance inspection, part 2 -Continued						
5251	Heater-cathode leakage current	-		---	---	---	---
5251	Grid No. 2 leakage current	-		---	---	---	---
(B) 1101	Secureness of base, cap, or insert	-		---	---	---	---
(B) 1111	Base pin solder depth	-		---	---	---	---
(B) 1105	Permanence of marking	-		---	---	---	---
	Quality conformance inspection, part 3						
---	Life test	-	Group C; Eb = 8,800 Vdc; Ec2 = 770 Vdc; Ib = 60 µAdc; t = 500 hours (minimum)	---	---	---	---
---	Life-test end points:						
5226	Line width A	-	Ib = 200 µAdc	Width	---	0.75	mm
5226	Line width C	-	Ib = 200 µAdc	Width	---	0.85	mm
5223	Modulation	-	Ib = 200 µAdc	Ecl	---	38	Vdc

NOTES:

1. When Ec2 is greater than 330 Vdc, Rg1 shall not exceed 0.5 Meg. When Ec2 is less than 330 Vdc, Rg1 shall not exceed 1.5 Meg.
2. A limiting aperture type of gun shall be used. The amount of limiting permitted shall be between 15 percent and 45 percent when Ib = 200 µAdc and the secondary electron current from the screen to the limiting aperture is zero.
3. The distance between the center of the unfocused, undeflected spot at low intensity (Ec1 near cutoff) and the center of the image of the masking aperture observed at high intensity of the unfocused, undeflected spot shall not exceed the limit specified. Ec1 should not be held at zero for more than approximately 30 seconds to prevent damage to the screen.
4. This test to be performed at the conclusion of the holding period.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Air Force - 17, 80
DSA - ES

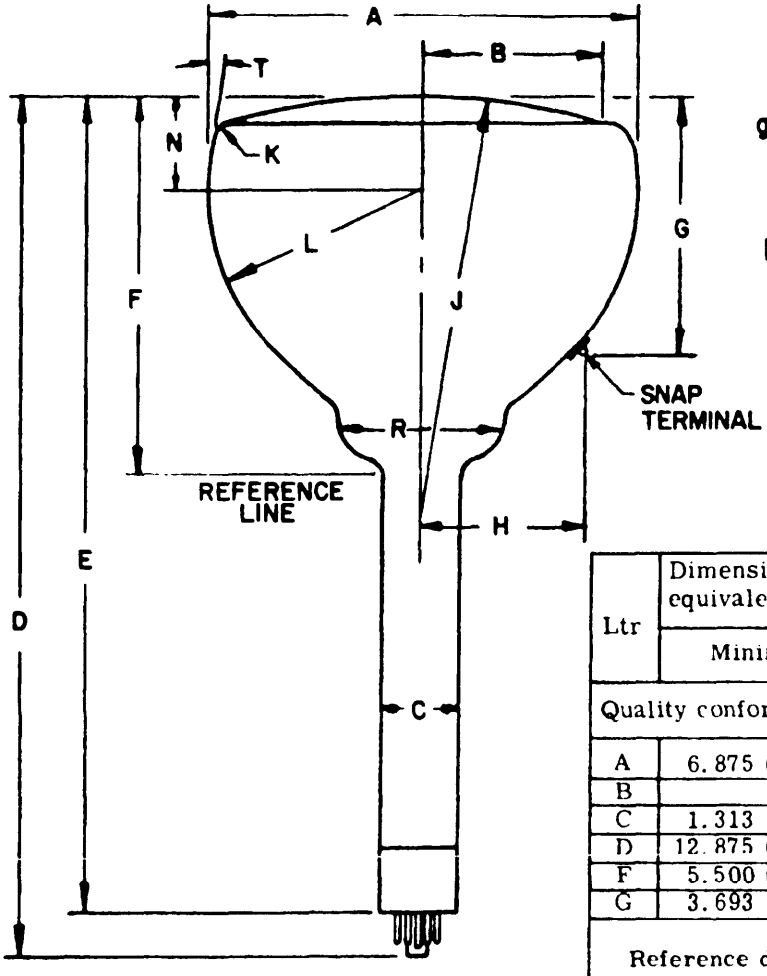
User activities:
Army - ME
Navy - AS, OS, MC, CG, SH
Air Force - 11

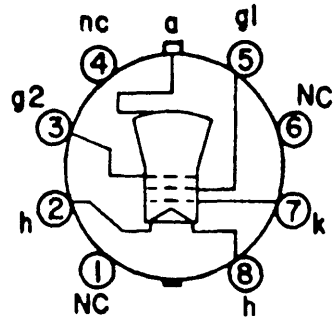
Preparing activity:
Navy - EC

Agent:
DSA - ES

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 2		
A	6.875 (174.63)	7.125 (180.98)
B	See note a	
C	1.313 (33.35)	1.427 (36.25)
D	12.875 (327.03)	13.625 (346.08)
F	5.500 (139.70)	6.000 (152.40)
G	3.693 (93.80)	4.193 (106.50)
Reference dimensions (see note e)		
E	12.687 (322.25)	
H	2.625 (66.68)	
J	24.000 (609.60) R	
K	.250 (6.35) R	
L	3.500 (88.90) R	
N	1.375 (34.93)	
R	3.125 (79.38)	
T	5°	

NOTES:

a. The minimum useful screen radius shall not be less than 3.00 inches (76.20 mm).
b. The bulb shall be a J56D or J56R type.
c. Applies to entire length of neck below reference line. The reference line is determined by point where gage 1.430 ±.003 (36.32 ±.08 mm) inside diameter and 2 inches (50.80 mm) in length will stop against bulb body.
d. The base shall be a medium shell octal 8-pin (B8-11) type or a medium long shell octal 8-pin type (this is identical with the medium shell octal 8-pin type except that the shell is 1.375 inches (34.93 mm) high nominal) or a small wafer octal with sleeve 8-pin (B8-26) type.
e. Reference dimensions are for information only and are not required for inspection purposes.

FIGURE 1. Outline drawing for electron tube type 7BP7A.

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